

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Intracellular	1			1		
	(b)	(i)		<i>Independent:</i> temperature (1) <i>Dependent:</i> time (taken) to reach zero absorbance (1)	2			2		2
		(ii)		Acclimatisation/ owtte reject stated temperature		1		1		1
		(iii)		A. {Colour change/ indicator} is dependent on a change in pH (1) B. (Hydrolysis produces) fatty acids {that lower the pH/ make it acid} (1) C. Buffer solution would {stabilise/ maintain/ control} pH (to pH10/ alkaline conditions) / 'mop-up' H ⁺ (1)		3		3		3

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	(c)	(i)		Using 18 seconds $2.78 \times 10^{-2} / 2.8 \times 10^{-2} / 2.63 \times 10^{-2} / 2.6 \times 10^{-2} = 3$ marks $0.0277777777777778 / 0.028 / 0.02631579 / 0.03 = 2$ marks (not standard form) $2.7 \times 10^{-2} = 2$ marks (incorrectly rounded) $2.78 \times 10^{-1} / 2.63 \times 10^{-1} = 2$ marks (incorrect conversion) Using 19 seconds $2.63 \times 10^{-2} / 2.6 \times 10^{-2} = 3$ marks $0.027 = 1$ mark (not standard form and incorrect rounding) NOT taking volume into account Using 18 seconds $1.39 \times 10^{-2} / 1.4 \times 10^{-2} = 2$ marks $1.38 \times 10^{-2} = 1$ mark (incorrectly rounded) $0.0139 / 0.014 = 1$ mark (not taking standard form into account) $1.3 \times 10^{-2} = 1$ mark (incorrectly rounded) Using 19 seconds $1.32 \times 10^{-2} / 1.3 \times 10^{-2} = 2$ marks $0.0132 / 0.013 = 1$ mark (not taking standard form into account) $1.31 \times 10^{-2} = 1$ mark (incorrect rounding) OR Allow $2 \times 0.25 = 0.5$ for 1 mark						
		(ii)		Cooling in the colorimeter/ Accuracy of equipment used to measure volumes/ not all volumes transferred (1)			1	1		1

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		(iii)		A. Inappropriate (temperature) range (1) B. {time taken/ graph} {is still decreasing/ has not levelled off}/ all enzymes still active (1) <i>Improvement :</i> C. Needs to test increase temperatures above 50°C, (to determine how well the enzyme is able to work at higher temperatures) (1) ignore any temperatures below 50°C D. smaller increments (to determine the optimum accurately) (1)			4	4		4
				Question 4 total	3	7	5	15	3	11